Marmara University Electrical-Electronics Eng. Dept.

CSE2003 Digital Design Laboratory Assignment #3

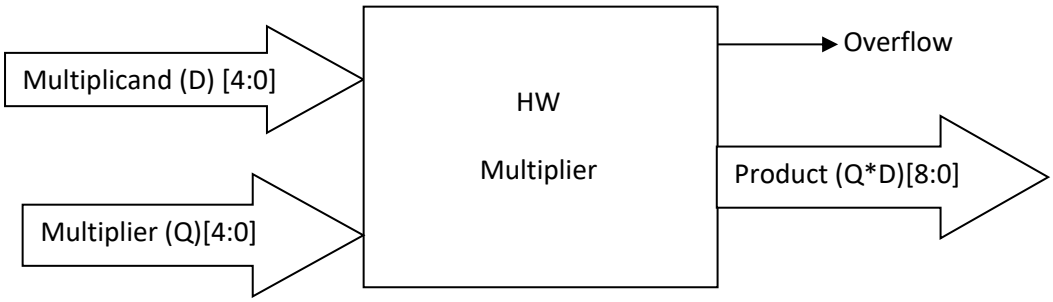
The studies will be uploaded to the system before due date (after two weeks). **Each group must present their own study. The group members have to work together and each member must know each step of the study.**

For this laboratory assignment, you will use Altera Quartus II software and DE-115 FPGA kit and design a hardware floating point multiplier that has the Input/output defiitions below.

The steps you need to follow: 1. You should design the circuit, **2.**write the verilog HDL code fort he designed circuit, **3.** Check if the designed hardware is correctly coded by using RTL viewer. Don't forget: your circuit may look different then yours since the compiler can make some optimizations. Please take this information into account when you compare the RTL viewer circuit and yours. **4.** Verify your design by using simulation tool (you need to check the inputs and outputs). **5.** implement and RUN your code using Altera DE-115 FPGA kit. Use the remote desktop connection (**Remote desktop connection ID**:256 521 711 **password**: dijitallab2021).

You need to upload the following files before due date:

1. A short video (<60 secs.) that shows all steps of the study. Especially, the implementation part must be included.
2. The report (in PDF format): 1. What is the purpose of the study? 2. What are the design details with necessary explenations. 3. The design and implementation steps (all steps with screenshots etc.). 4. The result and coclusion: did you reach the purposed result? If the answer is no, why? if it is yes, how?



All numbers in this assignment are in the IEEE 754 binary floating-point format and the inputs and output have 6 and 9 bits respectively. The FP format has three areas: the MSB represents the sign of the number (0: positive, 1: negative), the exponantial part and the mantissa.

The area lengths for the inputs:

S-1bit	E-2 bits	M-2 bits

The area lengths for the output:

S-1bit	E-3 bits	M-5 bits

The formula to convert the number from IEEE 754 format to decimal is given as:

$$\textit{For the numbers at input: } n_{input} = (-1)^{s} \cdot (1+M*2^{-2}) \cdot (2^{E-1})$$

$$\text{For the numbers at output: } n_{output} = (-1)^{s} \cdot (1+M*2^{-5}) \cdot (2^{E-3})$$

The formulas are different since area sızes are not same for input and output numbers.

Marmara University Electrical-Electronics Eng. Dept.

CSE2003 Digital Design Laboratory Assignment #3

The studies will be uploaded to the system before due date (after two weeks). **Each group must present their own study. The group members have to work together and each member must know each step of the study.**

Some examples:

Ex1: If the FP formatted number at **input** is: 01011 =>
The first bit is sign bit hence number is positive.

1. Exponantial area is E=10b. $2^{2-1}=2$
2. Mantissa, M=11b. $1+3*2^{-2}=1.75$
3. The number: + 2*1.75 = 3,5

Ex2: If the number at the **input** is 11101 =>
The first bit is sign bit hence number is negative.

1. Exponantial area is E=11b. $2^{3-1}=4$
2. Mantissa, M=01b. $1+1*2^{-2}=1.25$
3. The number: - 4*1.25 = -5

Ex3: If the number at the **output** is 111100011 =>
The first bit is sign bit hence number is negative.

1. Exponantial area is E=111b. $2^{7-3}=2^4$
2. Mantissa, M=00011b. $1+3*2^{-5}=1.09375$
3. The number: - 2^4*1.09375 = 17.5

Multiplication procedure for floating point numbers:
The floating point formatted inputs Q and D will be multiplied.

Step1: Same sign bits => positive result, otherwise => negative result.

Step2: normalize the exponentials by substracting 01b, then sum them. Add 01b to the sum. $E_{result}= E_D -1 + E_Q -1$.

Step3: multiply the mantissa areas as shown in the following formula: $M_{result} = 1.M_Q * 1.M_D$. In the formula, all the numbers are binary and the mantissa areas are used as fractional parts.

Possible results:

1) M_{result} = 1x.xxxx => move the radix point one place to the left and increase the exponent by 1. After operation, M_{result} = 1.xxxxx and $E_{result}= E_D -1 + E_Q -1 +1$.
2) M_{result} = 01.xxxx => no manipulation is needed.

Step4: E_{result} should be increased by 3 and cenverted to binary, use only fractional part of M_{result} (M=xxxxx) and use sign bit at MSB position:

Multiplication result in FP format: SEEEMMMMM

Marmara University Electrical-Electronics Eng. Dept.

CSE2003 Digital Design Laboratory Assignment #3

The studies will be uploaded to the system before due date (after two weeks). **Each group must present their own study. The group members have to work together and each member must know each step of the study.**

Multiplication Example:

Let us multiply the numbers in Ex1 and Ex2: (Q=01011)*(D=11101) = ?

Step1: sign bits are not same => resulted sign bit: 1

Step2: E_Q=10b=2, E_D=11b=3 => E_{result}= 3+2-2=3.

Step3: M_Q=11b and M_Q=01b =>

```
   1.11
   1.01
 *-------
    111
   000
  111
+-----------
  10.0011
```

M_{result}=10.0011 => the radix point should be moved to the one place left. Therefore E_{result} should be increased by one.

After Step3: E_{result}=3+1 = 4 and M_{result}=1.00011

Step4: increase E_{result} by 3 and convert it to binary: E_{result}=111. Use only fractional part for M_{result}=00011. The multiplication result in FP format: Product=111100011.

Overflows:

If the result's exponantial area exceeds 3 bits an overflow is occured. The result's Sign bit defines the type of overflow (positive or negative).

Overflow example:
11111*01111=?

Step1: result is neg., Step2: E=3-1+3-1=4,

Step3: 1.11*1.11=11.0001=> radix point should be moved to one place left and E should be increased by 1 => M=1.10001 and E=5.

Step4: E = 5+3=8=1000b and M=10001. The exponantial area has four bits=> overflow = 1, S=1, E=000 and M=10001. The result: Ov=1, Product=100010001.